Pixarch — Baseline Report

On-device agentic retrieval over personal data

31 October 2026

This report covers the **baseline**: the non-agentic, single-shot system that every later contribution is measured against. All numbers come from `docs/results/baseline_2026-10-31.md`, which `scripts/evaluate.py` regenerates from scratch.

> **Status: indicative, not publishable.** Relevance is single-judge and part of the corpus is generated. The numbers are good enough to catch regressions and to size the on-device problem; they are not good enough to support a claim.

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1 Labeled Dataset Creation

No dataset exists for mixed personal-archive search, so we built one. The corpus grew from 127 items to **629 manifest rows / 627 indexed** across **31 subtypes** — 14 document, 17 scene.

`data/raw/` is gitignored while `data/manifest.csv` is committed, carrying `source_url`, `license` and `attribution` per row, so the corpus is reproducible *by reference* rather than by redistributing images. `scripts/fetch_dataset.py` rebuilds it.

1.1 Sources

Bucket	Subtypes	Source	Licence
scene	beach, mountain, trees, vehicle, party, wedding, college, food, travel, fitness, mall, planes, outdoor_games, clothes, art, transit_hub	SUN397 (via HuggingFace)	research use
scene	xray	keremberke/chest-xray-classification	CC-BY-4.0
document	receipt	ICDAR-2019 SROIE	CC-BY-4.0
document	bill	katanaml-org/invoices-donut-data-v1	CC-BY-4.0
document	general_doc	FUNSD	research use
document	handwritten	IAM Handwriting Database	research use
document	book	Wikimedia Commons title pages	PD / CC
document	aadhaar, pan, dl, passport, credit_card, debit_card, passbook, bank_statement, report_card	**synthetic** (generated)	CC0

1.2 Why nine classes are synthetic

Wikimedia Commons holds **nine** usable Indian identity specimens in total — verified by search, not assumed. Beyond those, every image of an Aadhaar card, PAN card, passport, licence, bank passbook or payment card on the open web is a photograph of a **real person's** document, carrying their name, address, photograph and the number itself.

This is not merely a privacy preference; it is a property of the repository. `manifest.csv` is committed and stores `source_url`, and `fetch_dataset.py` reconstructs the corpus from it. Fetching such images would publish a permanent, public index of real people's identity documents and cause everyone who clones the repository to re-download them. India's Aadhaar Act 2016 additionally restricts the sharing of identity information. Building the evaluation set would be worse still: OCR ground truth would mean transcribing real Aadhaar numbers into a committed file.

Generation avoids all of this and produces *better* data: exact ground truth, balanced classes (21 PAN instead of 1), and controllable difficulty.

Safety properties of the generated values.

- Aadhaar numbers begin with **0 or 1** — UIDAI never issues those, so no generated number can collide with a real one.
- Card numbers are constructed to **fail the Luhn checksum**, so none can be a valid payment card.
- All person, bank and school names are fictional.

There is deliberately **no "SPECIMEN" watermark**. A token appearing in every synthetic document and no real one is a label leak straight into BM25 — the same defect as the filename leak that `schema.sql` documents removing (worth +0.125 p@1, measured, then deleted).

The nine real Commons specimens are kept at their original indices and excluded from the synthetic pool, so they remain a held-out realism check.

1.3 Handwriting

Every page-level handwriting dataset on HuggingFace is gated. Only line-level sets are reachable, and a raw IAM line is $\sim 2467 \times 128$ — a 19:1 strip that is neither a note a user would photograph nor anything CLIP can route sensibly. Pages are therefore **composed from real IAM handwriting lines**: the pixels are real, only the layout is ours, and the ground truth is the concatenated line text.

1.4 Ground truth and judgements

Artifact	Contents	Before	After
`ocr_groundtruth.jsonl`	character-exact + upstream transcriptions	20	**260**
`queries.jsonl`	≥ 3 per subtype, all 31 covered	20	**98**
`qrels.csv`	judgements, expanded by subtype	185	**2,183**

`scripts/make_qrels.py` now fails loudly if any corpus subtype has no query — the check that would have caught half the corpus becoming unevaluable when the subtype count went $15 \rightarrow 31$.

2 Models used for PC simulation

Role	Model	Footprint
Scene embedding + bucket routing	CLIP `ViT-B-32` / `laion2b_s34b_b79k`	**151.3 M params** — 605 MB fp32, 303 MB fp16, 151 MB int8
OCR	Tesseract 5, `eng+hin`	system package
Lexical index	SQLite FTS5 + BM25	in-index, no model

One CLIP instance serves **two** jobs — zero-shot bucket classification at ingestion and scene retrieval at query time — which halves what the on-device build must budget for.

Tesseract is the engine behind every number in this report. EasyOCR remains implemented behind the same protocol as a comparator but is not used here.

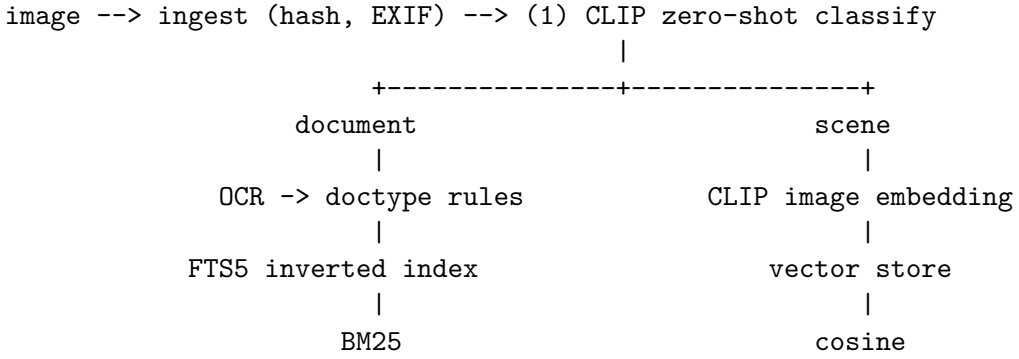
3 Finalized models for the edge device

Stage	Desktop	Android
OCR	Tesseract `eng+hin`	**ML Kit Text Recognition v2** (Latin + Devanagari)
Embedding	CLIP ViT-B-32 fp32	same weights, fp16 / int8
Index	SQLite FTS5	SQLite FTS5 (ships with Android)

All engines sit behind `OCREngine.extract(path) -> list[TextBlock]`, so the Android build swaps OCR by changing one configuration key. FTS5 was chosen partly because it is present on Android unchanged.

Budget implied by the measurements. CLIP at int8 is $\sim$151 MB resident, and the index projects to 85 MB for a 10,000-item gallery (§5.6). Ingestion is the real constraint: 1.56 hours for 10,000 items on a desktop GPU, which is why the architecture assumes heavy processing runs while the phone is idle and charging.

4 Baseline Architecture



At query time a **rule-first router** picks exactly one branch; CLIP prototype similarity is consulted only when the keyword rules abstain. Everything lands in a single `index.db`.

Deliberately absent. Each of the following is a later contribution that must not be pre-empted by the baseline it will be measured against: no score fusion, no agentic loop, no retry or clarification, no face branch.

Two choices the measurements justify.

- `doc_type` is its own FTS column **weighted** 5× against body text. Raw BM25 over OCR text is brittle when a scan is noisy and the one distinguishing phrase comes back mangled.
- **No filename column.** Indexing generated names leaked the class label into the ranking — worth +0.125 p@1, measured, then removed.

5 Results

Corpus 627 items · 98 queries · retrieval depth 107 · CLIP on CUDA · Tesseract.

5.1 Retrieval

System	p@1	p@5	MRR@10	nDCG@10	R-prec	recall@10
routed (baseline)	**0.773**	**0.781**	**0.803**	**0.773**	**0.693**	0.369
clip_only	0.629	0.596	0.644	0.580	0.478	0.278
bm25_only	0.361	0.355	0.378	0.349	0.312	0.157
filename	0.299	0.299	0.299	0.299	0.298	0.140

Routing beats both single-model comparators overall: **+0.144 p@1** over CLIP-only and **+0.412** over BM25-only.

Per bucket. Macro-averaged *within* each bucket over its queries, then macro-averaged *across* buckets so each counts once regardless of query count.

System	document p@1	scene p@1	macro p@1
routed	0.739	0.804	**0.772**
clip_only	0.370	**0.863**	0.616
bm25_only	**0.739**	0.020	0.379

The honest reading: routing's value is that it is *never catastrophic*. Each single-model system collapses on the bucket it was not built for — BM25 scores 0.020 on scenes, CLIP 0.370 on documents — while routing tracks the better of the two everywhere. On scenes alone CLIP-only is *better* than routed (0.863 vs 0.804), because every routing error is a pure loss.

5.2 Recall with ceilings

Recall@k cannot exceed $k/|\text{relevant}|$. With ∼20 items per subtype recall@10 caps at 0.48, so the raw figure badly understates performance.

System	recall@10	ceiling@10	% of achievable
routed	0.369	0.480	**77.0%**
clip_only	0.278	0.480	57.9%
bm25_only	0.157	0.480	32.7%
filename	0.140	0.480	29.2%

R-precision — self-normalising, with no ceiling artefact — tells the same story: **0.693** for routed against 0.478 for CLIP-only.

5.3 Module benchmarks

Bucket classification (CLIP zero-shot, ingestion). 618/627 = **98.6%**. Eight scene → document, one document → scene.

Query routing (rules, then CLIP). 95.9% — 4 of 98 misrouted. Three are the x-ray queries: the router has no radiography vocabulary, so CLIP decides, and a chest film reads as "a scanned document". The fourth is the intentionally unanswerable `q020`.

Document subtype tagging (`doctype.py`) — 70.7% over 287 documents. This module had never been measured.

Origin	n	Accuracy
synthetic	180	**100.0%**
fetched	107	**21.5%**

Misclassification	n	Cause
bill → receipt	20	invoices carry "total"/"invoice no"; no rule distinguishes them from receipts
book → unknown	20	`book-isbn` needs isbn/chapter/copyright; title pages have none
handwritten → unknown	20	no rule exists, and OCR on handwriting is weak
general_doc → unknown/receipt	20	generic forms have no distinguishing vocabulary
aadhaar/pan/passport → unknown	4	OCR failed to recover the key phrase

The 100% / 21.5% split is the headline: the rules are effectively *perfect on rendered text and poor on photographs*, so the aggregate 70.7% is an artefact of the synthetic:fetched ratio rather than a property of the module. `receipt` precision of 0.391 — recall 1.000, but 20 bills and 8 forms swept in — is the clearest single defect.

OCR accuracy (Tesseract, 258 items with ground truth).

Set	n	CER	WER
all	258	0.471	0.596
excluding `bill`	238	**0.420**	**0.527**
synthetic	180	0.418	0.507
fetched	78	0.592	0.801

`bill` is **not comparable**. The upstream invoice ground truth is a set of extracted *field values*, not a page transcription, so OCR that correctly reads the whole page is scored against a much shorter reference and its CER exceeds 1.0. This is a ground-truth mismatch, not an OCR failure.

The synthetic-versus-fetched gap (0.418 vs 0.592 CER) quantifies the domain gap the generated corpus carries: rendered text is roughly 30% easier than photographs and scans. The best subtype is `passbook` (0.187); the worst comparable one is `bank_statement` (0.655).

5.4 Query latency

System	Scope	n	mean	p50	p95	max
routed	all	98	8.0 ms	6.8	13.5	63.3
routed	document	49	**4.1 ms**	1.6	8.5	46.9
routed	scene	49	**11.9 ms**	12.4	15.2	63.3
bm25_only	—	98	0.6 ms	0.4	1.5	2.2

The branches are bimodal by an order of magnitude: BM25 is a SQL query (0.6 ms), while the scene branch pays a CLIP text-encoder forward pass (11.9 ms). Both sit far below interactive thresholds — **query time is not the on-device constraint**.

5.5 Ingestion throughput

Stage	Items	ms/image	images/sec
ingest	627	29.1	34.41
clip_encode	627	43.2	23.14
classify	627	26.9	37.22
embed_store	627	0.2	4209.58
ocr	393	**738.0**	**1.35**
end-to-end	627	**562.0**	**1.78**

Stages run over different denominators: `ocr` touches 393 items (287 documents plus 106 low-confidence items routed to both branches) while the rest touch all 627. A single ms/image averaged across stages would be meaningless.

Bucket	ms/image	images/sec
scene	99.4	10.06
document	1110.0	0.90

OCR is 82% of total indexing time and makes documents ∼11× more expensive than scenes, so mixed-gallery throughput depends entirely on the document:scene mix.

Per-item OCR is long-tailed — mean 691 ms, p50 478 ms, **p95 1567 ms**, **max 3048 ms** — so the mean understates the worst case by roughly 4×.

> **Projected: 1.56 hours to index a 10,000-item gallery** at this corpus's 46% document ratio (linear projection, desktop GPU).

5.6 Storage and memory

`index.db` is **5.3 MB** for 627 items.

Component	MB	Share
embeddings	2.58	48.3%
ocr_blocks (+ autoindex)	1.54	28.9%
doc_fields	0.39	7.4%
doc_fts (inverted index)	0.36	6.7%
items	0.25	4.7%

Measure	Bytes
per item, overall	8,525
per item, shared (row + CLIP vector)	5,840
extra per document (OCR + FTS)	5,866

Projected index for 10,000 items: 85 MB at this corpus's ratio; 58 MB all scene; 117 MB all document. With CLIP at int8 (∼151 MB resident), the whole system fits comfortably inside a mid-range phone's budget — the storage claim holds.

6 Repository

```
pip install -r requirements.txt
python scripts/fetch_dataset.py --per-class 20 --per-doc 20   # rebuild corpus
python scripts/make_qrels.py                                  # queries -> qrels
python scripts/build_index.py                                 # ingest + OCR + CLIP
python scripts/evaluate.py                                    # -> docs/results/
python app/demo.py                                            # Gradio UI
pytest                                                        # 94 tests
```

Path	Contents
`src/pixarch/{ingest,classify,ocr,doctype}.py`	ingestion pipeline
`src/pixarch/synthetic.py`	synthetic specimen generation
`src/pixarch/store/`	SQLite schema, FTS5, vector store
`src/pixarch/query/`	router, BM25, CLIP search
`src/pixarch/eval/`	metrics, module benchmarks, resource accounting
`configs/baseline.yaml`	every tunable
`docs/design-decisions.md`	why each choice, and the traps hit

7 Known limitations

1. **Relevance is single-judge** and declared by subtype rather than per item. The proposal calls for multi-judge agreement; that is outstanding.

2. **The `filename` comparator is inflated.** The fetcher names files after their class (`scn_beach_-020.jpg`), so path matching succeeds far more often than on a phone, where names look like `IMG_20231205_123456.jpg`. Treat 0.299 p@1 as a loose upper bound on the trivial floor, not the real floor.

3. **`doctype` rules do not cover `bill`, `book`, `handwritten` or `general_doc`** — 72 of 287 documents are mistagged. The rules were written when those classes had no data.

4. **No screenshots**, though the proposal names them as a gap in existing benchmarks (20–40% of a real gallery). Deferred.

5. **Synthetic documents are rendered, not photographed.** The 0.418 vs 0.592 CER gap bounds the optimism this introduces; the nine held-out real specimens are the check.

6. **X-ray queries misroute** and handwriting retrieval fails outright — both visible in §5.3, neither yet addressed.